

Beyond Pareto Fronts: A Leximax Universal-Regret Core for Robust Multicriteria Decision Support

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Abstract

Pareto efficiency is necessary but not sufficient for a unique recommendation. An efficient set tells a decision maker (DM) which alternatives are not obviously bad, but offers no principled way to choose one. Practice fills this gap with *post hoc* devices—weighted sums, TOPSIS, compromise programming, reference points, acceptability indices—each injecting implicit or stochastic preference judgements.

We propose the **Leximax Universal-Regret (LUR) core**, a leximin-refined preference-robust selection rule operating over a structured, declared, monotone scalarisation family. The DM defines an explicit family of rational questions (probes) and supplies normalisation bounds. LUR treats each probe's evaluation as an achievement function, calculating how far each alternative falls from the best attainable outcome under each probe, and sorts the resulting dimensionless regrets. The alternative whose worst performance is best is recommended, with an interpretable certificate listing the binding questions.

We prove existence and completeness, Pareto compatibility, stability of the tolerance-based leximax rule under bounded perturbations, and a probe perturbation bound with explicit constants. We also give an adaptive correlation-clustering construction empirically reducing, without eliminating, sensitivity to redundant criteria.

In a replicated benchmark and a frozen audit broadened protocol covering 11 methods, 10 held-out preference families, 8 front geometries, and m up to 20, LUR sits in the best-performing group on tail held-out loss. Its distinct empirical advantages over averaging methods are its interpretable certificate and reduced redundancy bias. Because the exact identity of the lexicographic minimiser can be highly sensitive to estimated normalisation bounds, the framework's primary robust deliverable is a *stability set* (or indifference class) evaluated over conservative interval bounds, with a unique point recommendation returning as a special case when verified margin conditions hold. We formulate an exploratory stochastic extension illustrated on a synthetic smart-grid economic dispatch model.

Keywords: Decision support, Multicriteria decision analysis, Robust decision making, Minimax regret, Achievement scalarising functions, Pareto efficiency

1. Introduction

1.1. The Pareto front as a safety filter, not a decision object

Since the work of Pareto, efficiency has been the cornerstone of multicriteria analysis: an alternative is Pareto efficient if no criterion can be improved without worsening another. This is a logically impeccable *safety property*—it removes alternatives that are dominated and therefore indefensible. It is, however, *decisionally incomplete*. For m conflicting criteria the efficient set is typically large or infinite, and the decision maker (DM) is left with a cloud of efficient alternatives and no principled rule for selecting one.

Confronted with this gap, practice resorts to *post hoc* selection devices, each of which reintroduces the very preference information the DM was unable to supply: a weighted sum requires weights and cannot reach non-convex parts of the front [17]; TOPSIS [12] requires a distance metric and weights; compromise

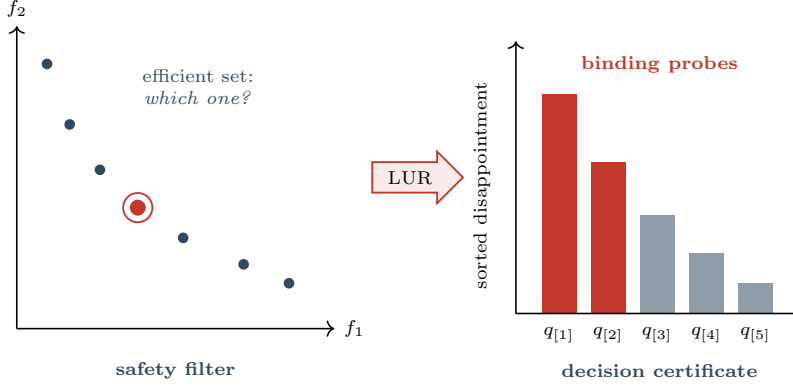


Figure 1: From a front to a certificate. The efficient set (left) removes dominated options but leaves the choice open. LUR returns one alternative together with its sorted regret certificate (right), whose largest entries—the *binding probes*—explain where the recommendation remains most fragile.

programming [32] requires a metric exponent and weights; knee-point rules [4] are scale sensitive; outranking methods such as ELECTRE [25] and PROMETHEE [5] require concordance/discordance thresholds; and hypervolume-based selection [33] favours extremes and is costly. These methods are valid and widely used, but they require the analyst or DM to commit to additional modelling choices—weights, metrics, thresholds, exponents, or reference points—that may be difficult to justify when preference information is intentionally incomplete. The fundamental issue is that **the Pareto front is a safety filter, not a complete decision rule.**

1.2. From a front to a certificate

We argue that, for the purpose of *choosing*, the efficient set is often insufficient as a final decision object. We propose supplementing it with a *regret certificate*: a vector of normalised disappointments over the rational questions the DM might reasonably ask about the criteria, sorted from worst to best. Rather than asking “which alternatives are efficient?” we ask

which alternative is least fragile across all declared rational interpretations of the criteria?

This turns the output of the analysis from a *set* into a *certified stability class*: a robust set of alternatives (often a singleton, if bounds are verified and margins hold) accompanied by an explanation of which questions remain most severely binding.

It is crucial to note what this approach does and does not do. LUR is not parameter-free; preferences are represented by an explicit auditable probe family, and it requires normalisation bounds and tolerance settings for numerical implementation. Its contribution is that these choices are declared, auditable, and separated from the final recommendation, not that they disappear. Figure 1 contrasts the efficient set as a safety filter with the regret certificate as the decision object.

1.3. The Leximax Universal-Regret core

We formalise this as the **Leximax Universal-Regret (LUR) core**. At its core, LUR is a *leximin-refined preference-robust selection rule* operating over a structured, declared, monotone scalarisation family. Given a feasible set and a *declared* class $\mathcal{Q}$ of admissible monotone probes (scalar “rational questions” over the normalised criteria), LUR treats each probe as an achievement function, ranks alternatives by the lexicographically sorted vector of their normalised disappointments across $\mathcal{Q}$, and returns the lexicographic minimiser together with its certificate. The construction rests on three principles: (i) *declared admissibility*— $\mathcal{Q}$ is part of the problem statement, not hidden inside a scalarisation; (ii) *Pareto compatibility*—under monotone probes a dominated alternative is never preferred to its dominator; and

(iii) *direct finite computation*—the method is primarily a *post hoc* robust selection rule operating exactly on a finite candidate set, though we also provide exact front-free formulations for structured continuous cases (§4).

1.4. Positioning and contribution

LUR belongs to *robust* multicriteria decision analysis. Its closest relatives are stochastic multicriteria acceptability analysis (SMAA) [15, 27], which integrates acceptability over a preference distribution; robust ordinal regression [10, 9], which reasons over the set of value functions compatible with elicited holistic judgements; and preference robust optimisation [1, 11], which optimises against an ambiguity set of utility functions. LUR differs from all three in object and output: it computes the *lexicographic minimax* (leximin) of normalised achievement shortfalls over an explicit, monotone probe family and returns an interpretable certificate; it requires no preference distribution, no elicited judgements, and no single scalarisation. We show that achievement scalarisation [30], the weighted sum, and the lexicographic maximin [21] are recovered as special cases of the probe family. Ultimately, LUR provides a concrete synthesis of these established aggregation mechanisms into a single, auditable robust selection rule equipped with a linear-size probe family.

Our contributions are:

- **Core theoretical contribution:** We define the LUR order and its auditable certificate, proving existence, completeness, Pareto compatibility *under an explicit separation condition*, dominated-solution exclusion, and stability of the tolerance-based rule under bounded perturbations (§3).
- **Computational contribution:** We present an exact candidate-set algorithm based on sequential lexicographic sorts, and outline a conditional formulation for structured continuous cases (§4).
- **Probe construction:** We propose an adaptive, redundancy-aware correlation-clustering probe family with empirical redundancy tests and a perturbation bound (§5).
- **Empirical validation:** We validate LUR on a broadened benchmark suite with held-out additive and non-additive preferences, and apply it to a stochastic smart-grid dispatch case study (§7).
- **Extensions:** We outline stochastic and multi-stakeholder variants as initial extensions demonstrating the framework’s flexibility (§6).

The remainder of the paper reviews related work (§2), develops the core and its guarantees (§3), the direct formulation (§4), the adaptive probes (§5) and the extensions (§6), reports the computational study (§7), discusses limitations (§8) and concludes (§9).

2. Background and related work

2.1. Pareto efficiency and post hoc selection

For a vector objective $F(x) = (f_1(x), \dots, f_m(x))$ (minimisation), x Pareto-dominates y if $f_i(x) \leq f_i(y)$ for all i with at least one strict inequality; the efficient set is the non-dominated set. Because it is typically large, a single recommendation is obtained by *post hoc* selection. The weighted sum, TOPSIS [12], compromise programming [32], knee selection [4], ELECTRE [24, 25], PROMETHEE [5] and hypervolume selection [33] are the standard tools; each requires weights, thresholds, a metric, or an already generated front. LUR does not eliminate modelling choices. It relocates them into a declared family of admissible monotone probes and, on finite candidate sets, computes the resulting recommendation exactly while returning a labelled regret certificate.

2.2. Achievement scalarising functions and ordered aggregation

Achievement scalarising functions (ASF) project a reference point onto the efficient set through a parameterised metric interpolating between L_1 and L_∞ [30, 18]. Ordered weighted averaging (OWA) [31] and its extensions (WOWA) [28] connect ordered aggregation to lexicographic maximin.

Remark 1 (Recovery of standard scalarisations). The LUR core recovers standard aggregation methods as exact special cases under specific, restricted probe families:

- **Weighted sum:** If $\mathcal{Q} = \{q_w(r) = w^\top r\}$, LUR exactly minimises the weighted sum of normalised criteria.
- **Chebyshev / ASF:** If $\mathcal{Q} = \{q_w(r) = \max_i w_i r_i\}$, LUR exactly minimises the Chebyshev distance to the ideal point.
- **Lexicographic minimax:** If $\mathcal{Q} = \{r_1, \dots, r_m\}$ (the singleton family), LUR performs a lexicographic minimax optimisation over the normalised regrets.

We do not claim LUR subsumes these methods with its full adaptive family—there it is a *different* aggregation paradigm. Its distinguishing features are that it sorts *normalised regrets* (dimensionless disappointments relative to each probe’s own attainable best and worst) over a *declared family of probes* and breaks ties through the whole sorted vector, and that it emits a certificate.

2.3. Robust MCDA: SMAA, robust ordinal regression, preference robust optimisation

The work closest to LUR lies in *robust* multicriteria decision analysis and minimax regret optimisation [14]. **SMAA** [15, 27] computes, by Monte-Carlo integration over a distribution of admissible weights (and possibly criterion values), acceptability indices describing how strongly each alternative is supported. **Robust ordinal regression** [10, 9] reasons over the entire set of additive value functions compatible with a DM’s holistic pairwise judgements, producing necessary and possible preference relations, sharing goals with inverse MCDA and the UTA family [13]. **Preference robust optimisation** [1, 11, 7, 29] optimises the worst-case expected utility over an ambiguity set of utility functions. LUR shares the robust philosophy—perform well across a whole class of admissible preferences rather than commit to one—but differs in three ways. First, its admissible class is an explicit family of *monotone probes*, not a weight distribution or an elicited polyhedron. Second, its criterion is the *lexicographic minimax of normalised regret*, not an expected acceptability, a necessary/possible relation, or a single worst-case utility. Third, its output is an interpretable *certificate* naming the binding coalitions. We note that SMAA is inherently an *exploratory* framework designed to visualise the robustness of alternatives across weight spaces, rather than a single-choice prescriptive rule. By forcing it to recommend the alternative with the highest first-rank acceptability index (ties broken by expected value), we create a comparable baseline for §7; however, we emphasise that LUR’s single-shot default recommendation is meant to *complement* rather than compete with SMAA’s rich exploratory indices. We likewise compare against a Savage minimax-regret (MMR) recommender over a sampled linear weight set (§7). More broadly, LUR is complementary to data-driven preference disaggregation and inverse MCDA [8], which infer preference parameters from holistic judgements: LUR instead dispenses with a fitted preference model and reasons over a declared class of questions, and could take an elicited probe family as input where such data exist.

2.4. Minimax regret and robust optimisation

Regret-based choice originates with Savage [26] and Loomes and Sugden [16]; minimax regret seeks alternatives minimising the largest shortfall from the best attainable value. Robust optimisation [2] and stochastic programming [3] address parameter uncertainty in the model rather than preference

uncertainty in the DM. LUR extends minimax regret in three directions: it is *lexicographic* (after the worst regret, minimise the second worst, and so on), *probe-based* (regret is computed over rational coalitions of criteria, not raw criteria), and *direct* (computed without enumerating the efficient set).

2.5. Social choice and fairness

Aggregating criteria resembles social choice, but treating criteria as voters invites Condorcet cycles; LUR avoids this by aggregating *regrets over declared questions*, not votes over criteria. For multi-stakeholder settings we draw on the maximin principle of Rawls [23] and the bargaining viewpoint of Nash [19]; we claim natural fairness properties (anonymity, monotonicity, worst-stakeholder protection), not a uniqueness theorem.

2.6. What is new relative to existing regret and robust-MCDA rules

While LUR shares motivation with several robust approaches, its mathematical object and output differ fundamentally:

- **Versus minimax regret:** LUR minimises a sorted vector of normalised probe regrets, not a single maximum regret over raw utilities or sampled weights.
- **Versus ASF/Chebyshev:** ASF commits to one scalarising function or one reference structure; LUR evaluates a declared family and certifies binding probes.
- **Versus SMAA:** SMAA integrates over a preference distribution and reports acceptability; LUR does not require a probability distribution and returns a leximax regret certificate.
- **Versus robust ordinal regression (ROR):** ROR reasons over compatible value functions after elicited comparisons; LUR can operate without holistic preference judgements.
- **Versus preference robust optimisation (PRO):** PRO optimises worst-case utility over an ambiguity set; LUR optimises lexicographically sorted normalised disappointments over interpretable monotone probes.
- **Versus ordered weighted aggregation (OWA):** LUR sorts regrets across declared questions, not criterion values themselves. OWA specifies an ordered weight vector, whereas LUR specifies a probe family; both therefore encode modelling choices in different forms. We include OWA utilities as a held-out evaluation family, not as evidence that LUR subsumes OWA.

The novelty is not minimax regret alone, but lexicographic minimax over normalised disappointments induced by a declared monotone probe family, with an auditable certificate and an adaptive redundancy-aware probe construction.

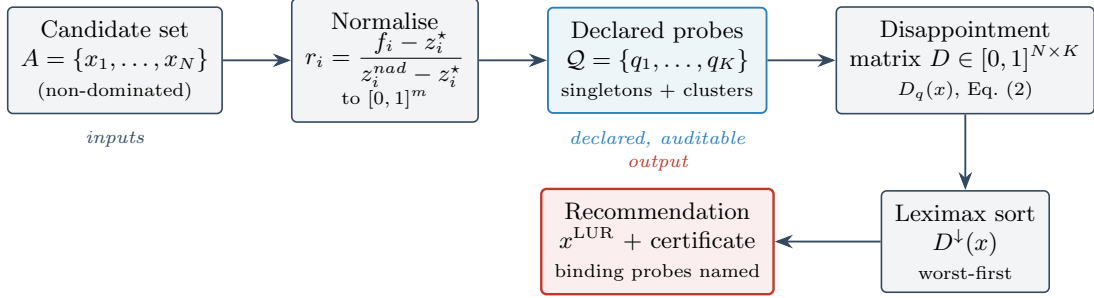
Table 1 summarises where LUR sits relative to representative MCDA and robust methods.

3. The Leximax Universal-Regret core

Figure 2 gives the end-to-end view of the rule developed in this section: from a candidate set, through normalisation and the declared probe family, to the leximax selection and its certificate.

Table 1: Comparison of LUR with representative selection and robust-MCDA methods.

Method	Main preference / model input	Requires candidate/front?	Single rec.?	Pareto comp.	Explanation / certificate	Main limitation
Weighted sum	Weight vector	No	Yes	Yes ($w > 0$)	Low	Cannot recover non-supported points in non-convex fronts
TOPSIS	Weights and distance metric	Yes	Yes	Not strictly	Low	Rank reversal; distance to non-feasible ideal
Compromise programming	Weights and L_p norm exponent	Yes	Yes	Yes ($p < \infty$)	Low	Non-linear scaling effects
ASF / Chebyshev	Reference point and scaling	No	Yes	Yes	Low	Requires setting an arbitrary reference point
MMR (sampled weights)	Linear weight distribution	Yes	Yes	Yes	Worst-case regret value	Limited to linear aggregations
SMAA	Distribution over preferences/weights	Usually	Not inherently (needs rule)	Usually	Acceptability indices	Exploratory; single-choice rule depends on post-processing
VIKOR	Weights and majority threshold	Yes	Yes	Not strictly	Low	Complex parameter interaction
Hypervolume contrib.	Reference point	Yes	Yes	Yes	Low	Favours extremes; sensitive to reference point
LUR (ours)	Declared monotone probe family; bounds; tolerance τ ; optional θ	Candidate-set exact; continuous cond.	Yes (or small class)	Yes (if separating probes)	Full labelled regret certificate	Sensitive to probe family, normalisation, and near-ties


 Figure 2: The LUR pipeline. Modelling choices are confined to the *declared* probe family $\mathcal{Q}$ and the normalisation bounds; the selection itself is an exact, deterministic leximax over the disappointment matrix, and the output is a single recommendation accompanied by an auditable regret certificate.

3.1. Problem setting and normalisation

Let $\mathcal{X} \subseteq \mathbb{R}^n$ be the feasible set and $F : \mathcal{X} \rightarrow \mathbb{R}^m$, $F(x) = (f_1(x), \dots, f_m(x))$, the vector objective.

Convention: All criteria f_i are assumed to be *minimised*. Maximisation objectives must have their sign reversed before normalisation. Consequently, lower normalised values r_i are better, and probes are non-decreasing functions of these costs.

For the finite case let $A = \{x_1, \dots, x_N\} \subseteq \mathcal{X}$ be a candidate set, e.g. an *a posteriori* approximation of the efficient set. Before evaluation, Pareto-dominated alternatives are removed from A unless the analyst explicitly requires selection over the raw un-filtered set. For the continuous case (§4) A is not required. Criteria are normalised to $[0, 1]$ by the positive affine map

$$r_i(x) = \frac{f_i(x) - z_i^*}{z_i^{\text{nad}} - z_i^*}, \quad z_i^* = \min_{x \in A} f_i(x), \quad z_i^{\text{nad}} = \max_{x \in A} f_i(x), \quad (1)$$

which attains both endpoints on A . A criterion is flagged as numerically degenerate if the range $z_i^{\text{nad}} - z_i^*$ approaches zero. Degenerate criteria are reported and excluded from non-singleton probe construction unless the analyst supplies external bounds.

For the finite case all extrema (z^* , z^{nad} , q^* , q^w below) are taken over the active candidate set A ; for the continuous case (§4) they are taken over $\mathcal{X}$. We use the upper attainable value over A in place of the true nadir over the efficient set; the two coincide when A contains the criterion-wise worst efficient

points, and otherwise the former is a conservative upper normalisation bound. The rule is invariant to positive affine rescaling of the criteria when the bounds are transformed consistently; it is not invariant to arbitrary monotone transformations (e.g. $f_i \mapsto \log f_i$), which change the distances that define regret—so such transformations are a modelling choice the analyst must make deliberately.

3.2. Admissible probes

A *probe* is a scalar function $q : [0, 1]^m \rightarrow \mathbb{R}$ encoding a rational decision question. The family $\mathcal{Q}$ is *declared*: it is part of the specification.

Definition 1 (Admissible probe). *A probe q is admissible if (i) it is non-decreasing in each r_i (monotonicity) and (ii) it can be rescaled so that $q(\mathbf{0}) = 0$, $q(\mathbf{1}) = 1$ (normalisation).*

Here $\mathbf{0}$ and $\mathbf{1}$ denote the all-zeros and all-ones vectors. The initial endpoint normalisation ($q(\mathbf{0}) = 0, q(\mathbf{1}) = 1$) serves to make values structurally scale-comparable across the heterogeneous family, before the empirical disappointment anchors (q^*, q^w) re-normalise the observed active range. Monotonicity gives *weak* Pareto compatibility; strict compatibility needs a separation property (Definition 2). Canonical admissible probes are the *singletons* $q_i(r) = r_i$; the *grand mean* $q_\mu(r) = \frac{1}{m} \sum_i r_i$ (the utilitarian question); the *grand maximum* $q_{\max}(r) = \max_i r_i$ (the egalitarian/worst-criterion question); and, for any coalition $S \subseteq \{1, \dots, m\}$, the coalition mean and maximum. All have non-negative weights or are maxima, hence are monotone.

Definition 2 (Separating family). *A probe family $\mathcal{Q}$ separates criteria if for every i there is a probe $q \in \mathcal{Q}$ strictly increasing in r_i . Including all m singleton probes $q_i(r) = r_i$ is sufficient; LUR always does so.*

3.3. Disappointment and regret

For each probe $q \in \mathcal{Q}$ let $q^* = \min_{x \in A} q(r(x))$ and $q^w = \max_{x \in A} q(r(x))$ (or a user-supplied worst-acceptable bound). Define the active range $R_q = q^w - q^*$. If $R_q < \varepsilon_q$ for a small numerical threshold, the probe is degenerate and is excluded from the primary decision vector. Otherwise, the *normalised disappointment* of x under q is

$$D_q(x) = \frac{q(r(x)) - q^*}{R_q} \in [0, 1], \quad (2)$$

so that $D_q(x) = 0$ iff x is q -optimal and 1 iff x attains the worst bound.

We use the term *disappointment* for the normalised shortfall of an alternative from the best attainable value of a single given probe, and *regret certificate* for the vector collecting these probe-wise disappointments. This is regret-like but differs from classical pairwise Savage regret unless the probe family is interpreted specifically as the uncertainty set of possible true preference functions.

3.4. The leximax order

Collect $D(x) = (D_{q_1}(x), \dots, D_{q_K}(x))$, $K = |\mathcal{Q}|$, and sort it in *descending* order to obtain $D^\downarrow(x) = (D_{[1]}(x) \geq \dots \geq D_{[K]}(x))$. The **Leximax Universal-Regret order** is

$$x \preceq_{\text{LUR}} y \iff D^\downarrow(x) \leq_{\text{lex}} D^\downarrow(y), \quad (3)$$

i.e. x is preferred if its worst disappointment is smaller; on ties, its second worst; and so on. The *LUR solution* is $x^{\text{LUR}} \in \arg \min_x^{\text{lex}} D^\downarrow(x)$, and its *certificate* is the labelled, sorted vector $D^\downarrow(x^{\text{LUR}})$ together with the probes attaining the largest entries. Figure 3 illustrates the comparison on a two-alternative example.



B is recommended: its worst disappointment is smallest. On a tie at the worst entry, the next sorted coordinate decides, and so on.

Figure 3: The lexicimax order. Each alternative’s probe disappointments are sorted worst-first; alternatives are then compared lexicographically on the sorted vectors. Here B wins because its largest disappointment (0.8) is below A ’s (0.9); had they tied, the second-largest entries would decide, and so on.

Exact LUR defines the mathematical complete preorder. In computation and practical reporting, a tolerance $\tau > 0$ is used only to define a numerical indifference class: we report alternatives whose sorted disappointment vectors are not separated from the exact winner by more than τ at the first differing coordinate. The τ -tolerance rule is not itself a transitive preorder, but rather a numerical reporting and stability device around the exact or margin-separated winner (Theorem 3).

3.5. Foundational guarantees

Theorem 1 (Existence and completeness of exact LUR). *For any finite A and finite $\mathcal{Q}$, x^{LUR} exists and exact $\preceq_{\text{LUR}}$ is a complete preorder on A .*

Proof. For finite A and finite $\mathcal{Q}$, each $D^\downarrow(x) \in [0, 1]^K$ is a well-defined real vector. The lexicographic order on $\mathbb{R}^K$ is reflexive, transitive and total, hence a complete preorder; since A is finite the lexicographic minimum exists. $\square$

Theorem 2 (Weak and strict Pareto compatibility). *Let x strictly Pareto-dominate y and let every $q \in \mathcal{Q}$ be monotone non-decreasing in each r_i . Then $D_q(x) \leq D_q(y)$ for all q (weak compatibility), so $x \preceq_{\text{LUR}} y$. If, in addition, $\mathcal{Q}$ separates criteria (Definition 2)—in particular if $\mathcal{Q}$ contains the singleton probes—then $D_q(x) < D_q(y)$ for some q and $x \prec_{\text{LUR}} y$.*

Proof. Let $x \prec_P y$, so $r_i(x) \leq r_i(y)$ for all i with strict inequality for some j . For any monotone non-decreasing probe q , $q(r(x)) \leq q(r(y))$; as q^*, q^w are constants independent of the alternative, $D_q(x) \leq D_q(y)$ for all q , giving $D^\downarrow(x) \leq D^\downarrow(y)$ componentwise after sorting and hence $x \preceq_{\text{LUR}} y$ (weak compatibility, monotonicity only). If $\mathcal{Q}$ separates criteria there is a probe q strictly increasing in r_j , for which $q(r(x)) < q(r(y))$ and thus $D_q(x) < D_q(y)$; the multiset of disappointments of x is then componentwise $\leq$ that of y with at least one strict drop, so the descending-sorted vectors satisfy $D^\downarrow(x) \leq_{\text{lex}} D^\downarrow(y)$ strictly and $x \prec_{\text{LUR}} y$. The counterexample in the main text shows the separation hypothesis cannot be dropped. $\square$

The separation hypothesis is necessary: with only the probe $q(r) = r_2$, the dominator $r(x) = (0, 1)$ and the dominated $r(y) = (0.1, 1)$ have equal disappointment, so monotonicity alone does not give a strict preference. LUR always includes the singletons, so it separates.

Corollary 1 (Dominated exclusion). *If x is Pareto-dominated by a feasible y and $\mathcal{Q}$ separates criteria, then x is not LUR-optimal.*

Proof. If x is dominated by a feasible y and $\mathcal{Q}$ separates criteria, Theorem 2 gives $y \prec_{\text{LUR}} x$, so x is not a lexicographic minimiser. Without separation only $y \preceq_{\text{LUR}} x$ holds and a dominated alternative may tie its dominator. $\square$

Theorem 3 (Stability of the implemented winner under a margin condition). *Let $\hat{D}_q(x)$ estimate $D_q(x)$ with $\eta = \max_{x,q} |\hat{D}_q(x) - D_q(x)|$. Suppose the exact winner x^{LUR} beats every rival y at the first coordinate $k(y)$ at which their sorted vectors differ by more than τ , with margin $D_{[k(y)]}(y) - D_{[k(y)]}(x^{\text{LUR}}) \geq \Delta$, and that all earlier coordinates agree within $\tau - 2\eta$. If $2\eta < \min\{\Delta - \tau, \tau\}$, the winner under $\hat{D}$ is unchanged.*

Proof. Write $\hat{D}_q(x) = D_q(x) + \xi_q(x)$, $|\xi_q(x)| \leq \eta$; sorting is 1-Lipschitz in $\|\cdot\|_\infty$, so each sorted coordinate moves by at most η . Fix a rival y and the deciding coordinate $k = k(y)$ with true margin $D_{[k]}(y) - D_{[k]}(x^{\text{LUR}}) \geq \Delta$. Under perturbation this margin is at least $\Delta - 2\eta$; since $2\eta < \Delta - \tau$ it stays above τ , so the τ -tolerance rule still decides coordinate k in favour of x^{LUR} . For the earlier coordinates $j < k$, which agree within $\tau - 2\eta$ by hypothesis, the perturbed gaps stay within τ and are treated as ties, so the decision is not pre-empted before k . As $2\eta < \tau$ no spurious strict difference is created earlier. Hence x^{LUR} still beats every rival and the winner is unchanged. (Exact comparison, $\tau = 0$, is excluded precisely because the second condition $2\eta < \tau$ then fails.) $\square$

Theorem 3 is deliberately stated for the tolerance rule: exact lexicographic comparison is *not* perturbation-stable, because a winner that beats a rival only at a later coordinate after an exact earlier tie can be reversed by an arbitrarily small perturbation of the earlier coordinate. The tolerance τ absorbs exactly those near-ties. We note that the margin condition Δ in Theorem 3 can be restrictive on densely packed candidate sets, where the separation between certificates may be very small. Without that margin the theorem gives no winner stability guarantee; the appropriate output is therefore a tolerance or stability set rather than a uniquely stable recommendation. For continuous $\mathcal{X}$, existence requires compactness of $\mathcal{X}$ and continuity of each $q \circ r$; uniqueness is not guaranteed and a small indifference class may arise.

4. Exact candidate-set computation and conditional direct formulations

LUR is primarily a candidate-set selection rule, and that is the setting of our experiments. We first present the exact finite algorithm. We then develop, more tentatively, how the recommendation can in principle be computed *directly* on a continuous feasible set, replacing the usual two-stage pipeline by a single robust search. We present the continuous case as a formulation with assumptions whose large-scale empirical validation we leave to future work.

4.1. Exact algorithm for finite candidate sets

The finite-set procedure is summarised in Algorithm 4.

The worst-case complexity is straightforward. With naive probe evaluation, winner selection costs $O(NKm + NK \log K + NK)$. Producing a complete lexicographic ranking of all candidates costs $O(NKm + NK \log K + NK \log N)$. Here $O(NKm)$ is the cost of probe evaluation and $O(NK \log K)$ is the cost of sorting the K -dimensional disappointment vector for each of the N candidates. The number of probes K is controlled by the construction of §5.

4.2. Cost of the anchors

Both the finite and continuous formulations need, for each probe, its attainable best $q^* = \min_x q(r(x))$ and worst $q^w = \max_x q(r(x))$. On a candidate set these are $O(N)$ scans; on a continuous $\mathcal{X}$ each is a scalar global optimisation, so the family contributes $2K$ such subproblems. Linear probes on a polyhedral feasible set yield linear anchor problems. For a general convex probe, minimising the probe may remain tractable, but maximising it over a convex set is not generally a convex optimisation problem. Non-convex feasible sets or probes may require global optimisation. Anchor computation can therefore be a substantial bottleneck and is not assumed cheaper than front approximation in all continuous cases.

Algorithm 1: Exact LUR recommendation on a finite candidate set

Input: Candidate set $A = \{x_1, \dots, x_N\}$; objectives $F(x)$; monotone probe family $\mathcal{Q} = \{q_1, \dots, q_K\}$; tolerance $\tau \geq 0$.

Output: LUR winner x^{LUR} and its certificate $D^\downarrow(x^{\text{LUR}})$.

1. Dominance filter: Remove any Pareto-dominated $x \in A$.

2. Normalisation: For each criterion i :

- Compute active range $z_i^* = \min_A f_i(x)$ and $z_i^{\text{nad}} = \max_A f_i(x)$.
- If non-degenerate, compute $r_i(x) = (f_i(x) - z_i^*) / (z_i^{\text{nad}} - z_i^*)$.

3. Anchor bounds: For each probe $q \in \mathcal{Q}$:

- Evaluate $V_{iq} = q(r(x_i))$ for all active $x_i \in A$.
- Find $q^* = \min_i V_{iq}$ and $q^w = \max_i V_{iq}$.

4. Regret evaluation: For each $x \in A$ and each valid probe $q \in \mathcal{Q}$:

- Compute $D_q(x) = (V_{iq} - q^*) / (q^w - q^*)$.

5. Certificate sorting: For each x , sort its regrets descending to get $D^\downarrow(x)$.

6. Selection: Return the lexicographically smallest row under τ -tolerance.

Figure 4: Finite-set LUR procedure.

4.3. Front-free formulation for structured cases

Lexicographic minimisation of the sorted vector over a continuous set $\mathcal{X}$ is exactly equivalent to sequentially minimising the cumulative sums of the p largest disappointments ($p = 1, \dots, K$), which admits an exact ordered-value linearisation [22]. Alternatively, the nested maximum can be handled via a single-stage heuristic surrogate using a strictly decreasing weight schedule $M^{-(k-1)}$, or via a smoothed log-sum-exp approximation for continuous gradients.

By formulating LUR directly on $\mathcal{X}$, a front-free computation is theoretically *available* for structured cases (e.g., linear multiobjective problems), as we demonstrate empirically in §7. However, because computing the per-probe anchor $q^w = \max_{x \in \mathcal{X}} q(r(x))$ requires maximising a convex function over a convex set, these $2K$ global anchor subproblems may be no cheaper than front generation in the general case. We are careful not to overclaim: the core contribution of this paper remains the finite-set *post hoc* selection rule, and we do not claim LUR as a general front-free mixed-integer solver.

5. Adaptive, redundancy-aware probe generation

5.1. The redundancy problem

The full coalition family (mean and maximum over every non-trivial subset of criteria) has size exponential in m . Worse, real criteria are often *redundant*: several measured objectives track the same underlying concern and are strongly positively correlated. Averaging selection rules implicitly *over-weight* such a group—it is counted once per member—biasing the recommendation.

Consider a 3-criteria numerical example where f_1 is a weak independent criterion and f_2, f_3 are redundant metrics (e.g. both measure environmental impact):

Alternative	r_1	r_2	r_3	Interpretation
A	0.70	0.20	0.21	weak independent criterion, good redundant group
B	0.45	0.35	0.36	balanced
C	0.25	0.55	0.56	good independent criterion, weak redundant group

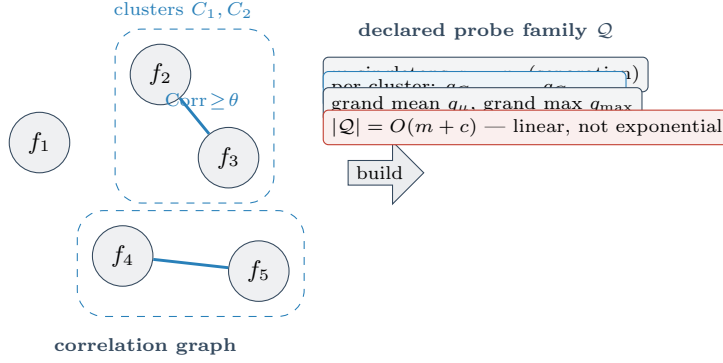


Figure 5: Adaptive probe construction. Positively correlated (redundant) criteria are grouped into clusters by thresholding the signed correlation at θ ; each cluster contributes a single mean and max probe instead of one per member. With the singletons (kept for separation) and the two grand probes, the family stays linear in m while curbing the over-weighting of redundant groups.

A full coalition family overcounts $\{r_2, r_3\}$ (which are highly correlated), implicitly double-weighting the environmental impact. The adaptive clustered family reduces the number of redundant coalition probes, but singleton probes remain for separation. Therefore, it reduces but does not eliminate redundancy sensitivity. We need a probe family that (i) is small, (ii) stays monotone, and (iii) is less sensitive to redundancy than averaging aggregation.

5.2. Correlation clustering

We group *redundant* criteria by single-linkage connected components of the graph whose edges join pairs with Pearson correlation $\text{Corr}(f_i, f_j) \geq \theta$. We explicitly state that both Pearson correlation and single-linkage clustering are deployed here as *empirical heuristics* rather than theoretically optimal redundancy measures. Pearson correlation captures linear dependence but is known to be sensitive to outliers; rank-based alternatives (e.g., Spearman) or information-theoretic measures could be substituted in future work. Similarly, single-linkage is chosen because it treats redundancy as connected substitutability, though it is notorious for “chaining” effects that can over-merge criteria. To mitigate this, we report empirical sensitivity to θ and compare it with complete-linkage in the supplementary material.

Redundancy is positive correlation: two criteria that rise and fall together carry duplicate information, whereas negatively correlated criteria are genuine trade-offs and must remain separate—hence we threshold the signed correlation, not its absolute value. This yields clusters $C_1, \dots, C_c$.

5.3. The adaptive probe family

LUR always includes the canonical anchors—the m singletons, the grand mean, and the grand maximum—and adds one mean and one maximum probe *per cluster* of size > 1 :

$$\mathcal{Q} = \{q_i\}_{i=1}^m \cup \{q_\mu, q_{\max}\} \cup \bigcup_{|C_k| > 1} \{q_{C_k, \text{mean}}, q_{C_k, \text{max}}\}. \quad (4)$$

Figure 5 illustrates the construction from the correlation graph to the resulting declared family.

A redundant group thus contributes *one* aggregate (cluster) question rather than a separate aggregate per member, so the family does not amplify a correlated group the way an averaging aggregate does, while the count stays $O(m + c)$ —linear, not exponential. We stress that this *reduces*, but does not eliminate, redundancy sensitivity: the singleton probes for the group’s members remain in $\mathcal{Q}$ (they are needed for separation, Theorem 2), so a redundant group can still influence lower-order lexicographic tie-breaking. The claim is therefore empirical—less redundancy bias than averaging methods on the

tested benchmarks (§7)—not a duplication-invariance theorem. All probes are non-negative-weighted sums or maxima, hence monotone, so Pareto compatibility (Theorem 2) is preserved.

Theorem 4 (Non-negative-probe Pareto compatibility). *If every probe in $\mathcal{Q}$ is a non-negative-weighted sum or a maximum of the normalised criteria, then LUR is Pareto compatible.*

Proof. If $q(r) = \sum_i w_i r_i$ with $w_i \geq 0$, then $r \leq r'$ componentwise implies $q(r) \leq q(r')$; likewise $q_{\max}(r) = \max_i w_i r_i \leq \max_i w_i r'_i = q_{\max}(r')$. All such probes are monotone non-decreasing, so Theorem 2 applies. $\square$

5.4. A perturbation bound (not a ranking guarantee)

We state the reduction effect as a regret-value perturbation lemma with explicit constants. We are deliberately modest: it bounds the change in individual disappointments, *not* the change in the LUR ranking or winner, and it depends on the normalisation floor ε , which is decision-critical.

Lemma 1 (Disappointment perturbation). *Suppose for a full-family probe q there is a reduced-family probe q' with $\sup_r |q(r) - q'(r)| \leq \eta$ (hence also $|q^* - q'^*| \leq \eta$ and $|q^w - q'^w| \leq \eta$), and let the normalisation ranges satisfy $q^w - q^* \geq \beta > 0$. Then for every x*

$$|D_q(x) - D_{q'}(x)| \leq \frac{4\eta}{\beta},$$

with the floor $\beta \geq \varepsilon$ in the worst case. The bound is vacuous when β is of order η , and informative only when the probe's attainable range is bounded away from zero—a regime the analyst can check directly.

Proof. Fix q and its approximant q' with $\sup_r |q(r) - q'(r)| \leq \eta$; then also $|q^* - q'^*| \leq \eta$ and $|q^w - q'^w| \leq \eta$. Write the ranges $N = q^w - q^*$ and $N' = q'^w - q'^*$, both $\geq \beta > 0$. Using $D_q(x) \in [0, 1]$,

$$|D_q(x) - D_{q'}(x)| = \left| \frac{q(x) - q^*}{N} - \frac{q'(x) - q'^*}{N'} \right| \leq \frac{|q(x) - q'(x)| + |q^* - q'^*|}{N'} + D_q(x) \frac{|N - N'|}{N'} \leq \frac{2\eta}{\beta} + 1 \cdot \frac{2\eta}{\beta} = \frac{4\eta}{\beta}.$$

The constant is explicit and the dependence on the range floor β (worst case ε) is exhibited, making clear when the bound is informative. $\square$

We emphasise that Lemma 1 provides a perturbation bound on the regrets, but it does *not* guarantee exact rank preservation of the alternatives. Thus, the probe reduction relies on theoretical bounds for safety but is ultimately an empirical heuristic whose efficacy is demonstrated in §7.

Whether the reduced family admits such an η -close q' for every full-family probe is *not* established by correlation clustering in general; we therefore present Lemma 1 as a sensitivity statement and rely on the empirical redundancy study (§7) for evidence that clustering retains decision quality.

Remark 2 (Residual redundancy sensitivity). Clustering does not establish duplication invariance. Singleton probes are retained for separation, so duplicating a criterion also duplicates coordinates in the sorted disappointment vector and can affect later lexicographic comparisons. Cluster-level probes reduce the empirical influence of correlated criteria, but the remaining sensitivity must be measured rather than assumed to vanish, even under perfect correlation.

5.5. Empirical reduction

Table 2 (plotted in Figure 6) reports the realised reduction on concave instances. The full family here is $\{q_{\text{mean},S}, q_{\text{max},S} : \emptyset \neq S \subseteq \{1, \dots, m\}\}$, of size $2(2^m - 1) - m$ (mean and max coincide on singletons); the adaptive family is linear in m , giving a nearly four-thousand -fold reduction by $m = 15$ with, as §7 shows, little loss of decision quality.

Table 2: Probe-family size: full coalition family $(2(2^m - 1) - m)$ probes) vs. adaptive correlation-clustering family (concave instances, $N = 400$).

m	full coalitions	adaptive probes	reduction
3	11	5	$2.2\times$
5	57	7	$8.1\times$
8	502	10	$50.2\times$
10	2,036	12	$169.7\times$
15	65,519	17	$3,854.1\times$

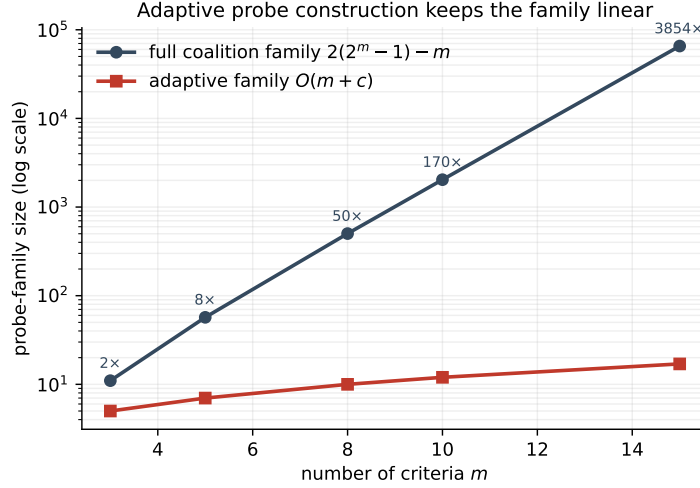


Figure 6: Probe-family size versus the number of criteria m (log scale). The full coalition family grows as $2(2^m - 1) - m$, whereas the adaptive correlation-clustering family stays $O(m + c)$, giving a nearly four-thousand-fold reduction by $m = 15$ with little loss of decision quality (§7).

5.6. Greedy probe selection (optional)

When even the adaptive family is larger than needed, a greedy set-cover heuristic selects a minimal subfamily distinguishing all dominant pairs; for the max-coverage variant it attains a $(1 - 1/e)$ guarantee and for full coverage a $1 + \ln d$ guarantee [20, 6]. We do not use it in the experiments, which already run on the compact adaptive family.

6. Extension: stochastic regret certificates

When criteria are noisy, model $f_i(x)$ as drawn from a joint distribution $P(x)$. Each probe value is then a random variable; for non-linear probes (e.g. $q_{\max}$) its mean $\mu_q(x)$ is *not* a simple function of the criterion marginal means. Let ρ be a generic monotone risk functional over the scenarios, for example the sample average $\rho_q(x) = \frac{1}{S} \sum_{s=1}^S q(r(x, \omega^{(s)}))$ or the empirical Conditional Value-at-Risk $\rho_q(x) = \text{CVaR}_\alpha(q(r(x, \omega)))$. The normalisation anchors must be taken consistently on this functional: $\rho_q^* = \min_{x \in A} \rho_q(x)$ and $\rho_q^w = \max_{x \in A} \rho_q(x)$. We define the *stochastic regret* as

$$D_q^\rho(x) = \frac{\rho_q(x) - \rho_q^*}{\rho_q^w - \rho_q^*}, \quad (5)$$

assuming the active range is non-degenerate ($\geq \varepsilon_q$). Using CVaR_α penalises both expected disappointment and its sample uncertainty. Because CVaR is a monotone risk measure, exact Pareto compatibility is preserved.

Proposition 1 (Scenario-wise Pareto compatibility). *For finite A , finite $\mathcal{Q}$, and sample-average probe means $\hat{\mu}_q(x) = \frac{1}{S} \sum_s q(r(x, \omega^{(s)}))$, if x weakly dominates y in every scenario and strictly dominates*

it in at least one criterion for at least one scenario, then the sample-average LUR rule does not prefer y to x . If $\mathcal{Q}$ separates criteria and the dominance is strict on a separated coordinate, then x is strictly preferred by the exact LUR order. We caution that this requires strict stochastic dominance across all scenarios, which is a very strong assumption as real stochastic problems often exhibit crossing dominance (where an alternative dominates in one scenario but is dominated in another).

Proof. If x weakly dominates y under all scenarios and strictly dominates it in at least one criterion for at least one scenario, then $r(x, \omega^{(s)}) \leq r(y, \omega^{(s)})$ for all s , with strict inequality on at least one coordinate for some scenario. For any monotone probe q , $q(r(x, \omega^{(s)})) \leq q(r(y, \omega^{(s)}))$ for all scenarios. The functional $\rho_q(x) = \frac{1}{S} \sum_s q(r(x, \omega^{(s)}))$ preserves this weak inequality, meaning weak Pareto compatibility is preserved. If $\mathcal{Q}$ separates criteria and dominance is strict on a separated coordinate, then $\rho_q(x) < \rho_q(y)$ for that probe. Since the normalisation constants ρ_q^* and ρ_q^w are fixed across alternatives, the corresponding regret is strictly lower for x , giving $x \prec_{\text{LUR}} y$ by the logic of Theorem 2. $\square$

Proposition 2 (Sample-average consistency under a margin condition). *Let A and $\mathcal{Q}$ be finite. For each $x \in A$ and $q \in \mathcal{Q}$, suppose $q(r(x, \omega))$ is i.i.d. across scenarios and has finite variance. Let $\mu_q(x) = \mathbb{E}[q(r(x, \omega))]$, and let $\hat{\mu}_q(x)$ be its sample average. Define population and sample disappointment vectors using their corresponding anchors. If the population tolerance-LUR winner satisfies the margin condition of Theorem 3, then the sample-average LUR winner converges in probability to the population winner as $S \rightarrow \infty$.*

Proof. Since A and $\mathcal{Q}$ are finite, the weak law of large numbers gives uniform convergence in probability of $\hat{\mu}_q(x)$ to $\mu_q(x)$. Because the minima and maxima over finite sets are continuous operations, the estimated anchors $\hat{q}^*$ and $\hat{q}^w$ converge in probability to the population anchors. Therefore, the sample disappointment vectors converge uniformly in probability to the population disappointment vectors. Under the margin condition of Theorem 3, there exists an $\eta > 0$ such that perturbations strictly within η do not change the tolerance-LUR winner. Since the probability of the maximum estimation error exceeding η vanishes as $S \rightarrow \infty$, the sample-average LUR winner converges in probability to the population winner. $\square$

§7 gives an empirical confirmation.

7. Computational study

7.1. Design and protocol

Software Architecture and Reproducibility. All experiments are implemented in the accompanying modular Python reproducibility package. Core selection methods (`lur.methods`), performance metrics (`lur.metrics`), and test problems (`lur.problems`) are separated. A recommendation and its labelled disappointment certificate are obtained directly from an objective matrix F :

```
from lur.methods import lur
winner, D, labels, _ = lur(F, return_detail=True)
certificate = sorted(zip(labels, D[winner]),
                    key=lambda item: item[1], reverse=True)
```

The protocol uses `pyarrow` for Parquet persistence and records the configuration, scientific-source, and environment identity of each run. A single command executes the frozen pipeline; failed and missing gates remain in the generated report.

Section 7.2 reports the main robust benchmark (7,200 instances) using this frozen broadened audit protocol. Sections 7.3–7.9 report the mechanistic analysis on a controlled 480-instance study used to understand behaviour, redundancy, and sensitivity.

We evaluate selection *quality* under a held-out protocol: each method picks one alternative using only its own machinery, and the pick is scored against preference models that *no* method used during selection. To avoid biasing the evaluation toward the additive/scalarising logic of the probes, the held-out set spans **six** families—random linear $-w^\top r$, weighted Chebyshev $-\max_i w_i r_i$, augmented ASF, CES, a *non-additive* 2-additive Choquet integral with random non-negative Möbius masses on singletons and pairs, and a *satisficing* threshold utility penalising exceedances of random aspiration levels. Weights are drawn from a uniform simplex; 300 utilities are sampled per family per replication. We report two non-negative, lower-is-better losses, each normalised per utility against the best and worst feasible alternatives: the *mean held-out loss* and the *tail held-out loss*. CVaR_{10%} (the average loss over the worst 10% of utility draws) serves as our robust tail measure. We intentionally report both mean loss and tail loss, emphasizing that robust methods will typically underperform on average in exchange for bounding the worst-case disappointment.

Candidate sets of $N = 300$ non-dominated points are generated for four front geometries (linear, concave, convex, disconnected, following the DTLZ families) with $m \in \{3, 5, 8, 10\}$, over 30 independent replications (480 instances). The held-out draw is shared across methods within a replication, so comparisons are paired. **Baselines and parameters.** TOPSIS, compromise programming (CP, $p=2$), knee selection (distance-to-hyperplane), achievement scalarisation (ASF, reference at the ideal), and a random-weight popularity rule (RW) all use equal weights where applicable. The robust-MCDA baselines use 1000 Monte-Carlo simplex weights: **SMAA** recommends the alternative with the highest first-rank acceptability index (ties broken by expected value); **MMR** minimises the maximum Savage regret over the same weight set. LUR uses the adaptive probe family with $\theta = 0.6$ and the tolerance-based lexic-max rule. Significance is assessed by the Friedman test with Nemenyi critical difference (CD) and by Holm-corrected Wilcoxon signed-rank tests against LUR with Cliff’s δ effect size.

7.2. Main Benchmark: The Broadened Protocol (7,200 instances)

To guard against confirmation bias we additionally ran a frozen audit protocol (frozen config, single-command reproduction) on a deliberately broadened suite: **11** methods (adding VIKOR, hypervolume contribution, and Chebyshev minimax regret), **ten** held-out families spanning additive *and* non-additive preferences (adding sparse-linear, L_p , OWA, Choquet and satisficing), Dirichlet weight shapes $\alpha \in \{0.2, 1, 5\}$, eight geometries and m up to 20 (7,200 paired instances). The protocol fixes, in advance, a set of *acceptance gates*, each tied to one precise claim; we report the outcome verbatim in Table 3 rather than only favourable numbers.

On this harder suite LUR has tail-loss rank 4.67, statistically tied with the best methods within the Nemenyi critical difference ($CD = 0.178$): HV 4.27, ASF 4.65, MMR 4.65, LUR 4.67, VIKOR 4.71, ChebMMR 4.99, then CP 5.32, TOPSIS 6.98, knee 7.21, RW 9.26 and SMAA 9.30 (Figure 7). A frozen *non-inferiority* test (paired bootstrap, margin 0.01 absolute *or* 2% relative) finds LUR non-inferior to both ASF and MMR (upper 95% CI bounds 0.0015 and 0.0021), with negligible effect size (Cliff’s $\delta \leq 0.015$)—supporting “practically equivalent to the best robust scalarisations” rather than “better”. Direct computation is demonstrated on a continuous linear MOP, where LUR via sequential LPs uses 19 solver calls and *no* front enumeration vs. 150 for generate-then-select, at equal or better tail quality. On a secondary MILP test, direct computation resolves the exact answer in just 4 solver calls compared to 60 for exhaustive generation. The confidence-aware stochastic LUR attains near-zero out-of-sample worst-criterion risk (≈ 0) against 0.28 for TOPSIS and 0.52 for SMAA, and the Rawlsian variant achieves the

Table 3: Frozen audit acceptance gates (reproduced at full scale). 9 PASS, 1 CHECK.

Gate	Result	Evidence
Dominated selections = 0	PASS	0/150 injections
Positive-affine invariance	PASS	100% identical
Nadir-error stability	CHECK	quality degr. 0.1533 (thr. 0.05)
Non-inferior vs ASF (tail)	PASS	CI upper 0.0015 < 2% margin
Non-inferior vs MMR (tail)	PASS	CI upper 0.0021 < 0.01 margin
Beats practical baselines (tail)	PASS	δ : TOPSIS .17, CP .07, SMAA .61
Redundancy < averaging (grouped)	PASS	0.256 vs 0.359
Adaptive $\approx$ full (tail gap ≤ 0.01)	PASS	gap 0.0053
Direct computation (LP + MILP)	PASS	19 vs 150; 4 vs 60 solver calls
Stochastic LUR does not hurt tail loss	PASS	risk 0.383 vs 0.383

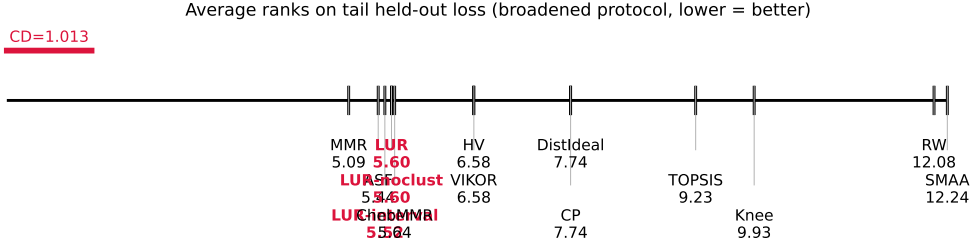


Figure 7: Average ranks on tail held-out loss over the broadened protocol suite (11 methods, 10 families, 8 geometries, 7,200 instances; Nemenyi CD = 0.178). LUR sits in the best-performing group on worst-case tail loss while providing an auditable certificate.

lowest worst-stakeholder regret (0.362) and inequality (Gini 0.207) among average-, Nash- and max-min rules.

The single CHECK outcome on Gate 3 (Nadir-error stability) serves as an important practical flag. It indicates that LUR’s recommendation can be sensitive to incorrect point estimates of the nadir. For practitioners, this CHECK explicitly means that if the true nadir is unknown or highly uncertain, they should not rely on naive point estimates, but must instead supply conservative interval bounds (evaluating worst-case regret over a box of plausible nadirs) to ensure true robustness, as further discussed in §8.

7.3. Mechanistic Analysis: The Controlled 480-instance Study

Table 4 reports mean and tail loss averaged over all geometries and m . On *tail* loss the four robust/compromise methods MMR (0.470), ASF (0.472), LUR (0.481) and CP (0.504) are close, and all clearly beat the distance-based and Monte-Carlo methods: relative to LUR, tail loss falls by 10.8% vs TOPSIS (0.539), 20.6% vs knee (0.606), 46.6% vs RW (0.900) and 47.4% vs SMAA (0.914). Figure 9 shows the average-rank CD diagram (Friedman $p < 10^{-15}$, $n = 480$, CD = 0.48). Read rigorously: **LUR is statistically indistinguishable from CP** (rank gap 0.02) and **very slightly behind the two best methods MMR and ASF** (gaps 0.59 and 0.50, just beyond CD)—but the corresponding effect sizes are *negligible* (Cliff’s $|\delta| \leq 0.034$). Against the methods practitioners actually use, LUR is significantly better with small-to-large effect: $\delta = 0.087$ (TOPSIS), 0.285 (knee), 0.850 (RW), 0.873 (SMAA), all $p < 10^{-4}$. LUR is clearly ahead of the default distance-based methods and of the SMAA-derived first-rank single-choice recommender used in this benchmark. We stress again that SMAA’s primary value is exploratory and visual; rather than claiming superiority over SMAA, we frame LUR’s single-shot default recommendation as strictly complementary to SMAA’s rich weight-space indices. We therefore claim only that *LUR is in the best-performing group on worst-case loss and far better than*

Table 4: Held-out loss over four geometries $\times m \in \{3, 5, 8, 10\}$, 30 replications (480 instances; lower is better). Tail loss is the worst-case ($\text{CVaR}_{10\%}$) metric. Best in each column in **bold**; on tail loss LUR is within a negligible effect size ($|\delta| \leq 0.034$) of MMR/ASF/CP.

Method	mean loss	tail loss
TOPSIS	0.220	0.539
CP	0.207	0.504
Knee	0.308	0.606
RW	0.462	0.900
ASF	0.234	0.472
SMAA	0.468	0.914
MMR	0.228	0.470
LUR	0.245	0.481

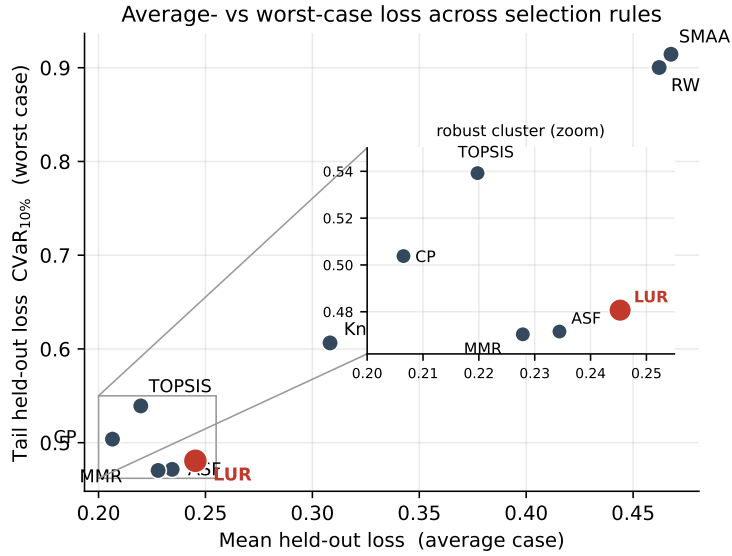


Figure 8: Average- versus worst-case held-out loss across selection rules (480-instance mechanistic study; lower-left is better). LUR sits in the robust cluster with ASF, MMR and CP on worst-case (tail) loss while the averaging and Monte-Carlo rules (RW, SMAA) trade away worst-case protection. The inset zooms into the robust cluster.

averaging/Monte-Carlo selection, not that it dominates the strongest robust scalarisations. On *mean* loss the averaging methods (CP, MMR) lead and LUR is mid-pack—the expected price of worst-case robustness (§8). Figure 8 makes this average-versus-worst-case structure explicit: LUR lies in the robust lower-left cluster, well separated from the averaging/Monte-Carlo rules.

7.4. Behaviour across preference families

Table 5 (visualised as a heatmap in Figure 10) breaks the loss down by held-out family on a representative setting (concave, $m = 8$), including the two non-additive families. The structure is a clear trade-off: averaging methods (TOPSIS, SMAA) are strong on linear utilities but weak on the max-type families; ASF and MMR are the reverse—best on Chebyshev/CES but worst on linear. LUR sits between the two camps: it removes much of ASF’s linear-utility weakness (0.598 vs 0.674) while staying competitive on the max-type *and* the non-additive Choquet (0.368) and satisficing (0.468) families. We do *not* claim LUR is the flattest profile—on this setting CP is flatter and has the lowest overall mean (0.404)—only that LUR is robust across additive and non-additive preferences without being tuned to any.

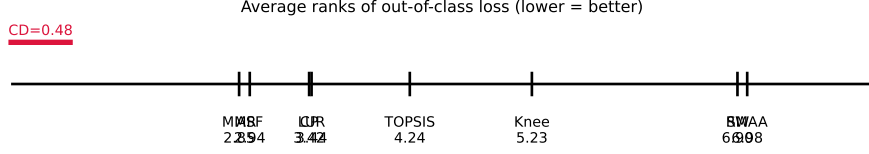


Figure 9: Average ranks on tail (worst-case) held-out loss with Nemenyi critical difference ($n = 480$, $\alpha = 0.05$, $CD = 0.48$). LUR groups with the best methods (MMR, ASF, CP) and outperforms the SMAA-derived first-rank single-choice rule used in this benchmark; the rank correlations with CP and ASF are extremely high (> 0.95).

Table 5: Held-out loss by preference family (concave, $m = 8$, 30 replications), including the non-additive Choquet and satisficing families. Lowest per column in **bold**. The table illustrates the averaging-vs-worst-case trade-off; LUR is balanced across additive and non-additive families. The *augmented ASF* family is the weighted Chebyshev term plus a small (0.05) linear augmentation on the same weights, so it induces almost the same induced ranking—and hence almost the same held-out loss—as the Chebyshev column; the two coincide to three decimals except where the augmentation flips a ranking (e.g. MMR, 0.442 vs 0.443).

Method	linear	Cheby.	aug. ASF	CES	Choquet	satisfice	overall
TOPSIS	0.391	0.511	0.511	0.563	0.341	0.511	0.471
CP	0.486	0.458	0.458	0.440	0.307	0.278	0.404
Knee	0.477	0.503	0.503	0.524	0.479	0.570	0.509
RW	0.403	0.519	0.519	0.569	0.379	0.552	0.490
ASF	0.674	0.434	0.434	0.329	0.288	0.388	0.425
SMAA	0.398	0.521	0.521	0.575	0.367	0.546	0.488
MMR	0.653	0.442	0.443	0.348	0.344	0.402	0.439
LUR	0.598	0.454	0.454	0.374	0.368	0.468	0.453

7.5. Certificate regret and uniformity

By construction LUR minimises the leximax of its certificate regret. Empirically (concave, $m = 8$) LUR attains worst-case certificate regret 0.708, far below every alternative (MMR 0.921, CP 0.936, ASF 0.953, TOPSIS 0.974), with one of the most uniform regret profiles (sorted-disappointment std 0.219, second to MMR’s 0.207). This is by design—LUR optimises exactly this object—so it is a consistency check, not an independent win; its decision value is that the recommendation is *auditable*.

7.6. Robustness to redundant criteria

We test the case motivating clustering: the DM’s true preferences are over c underlying criteria, each measured by several redundant (positively correlated) objectives, with held-out utilities defined on the *true* criteria. The clustering recovers the redundancy groups essentially exactly. Table 6 and Figure 11 (four group structures $\times$ three geometries, 30 replications) show LUR tied with ASF for best grouped loss and in the best group on tail loss; the averaging and Monte-Carlo methods, which amplify redundant groups, are 19–35% worse on grouped loss (TOPSIS 0.332, SMAA 0.413 vs LUR 0.268). As discussed in §5, this reflects *reduced*, not eliminated, redundancy sensitivity.

7.7. Distinctness, dominated exclusion, ablation, sensitivity

LUR and the SMAA recommender coincide on only 1.2% of instances and select alternatives a mean normalised distance 0.86 apart: LUR is operationally a different rule, not a re-description of acceptability analysis. Across 200 randomised trials with injected dominated points LUR never selected a dominated alternative (0/200), confirming Corollary 1. Ablation (concave, $m = 8$): dropping the singletons raises loss (0.456 $\rightarrow$ 0.483) and worst-case certificate regret (0.701 $\rightarrow$ 0.783); a max-only family lowers mean loss

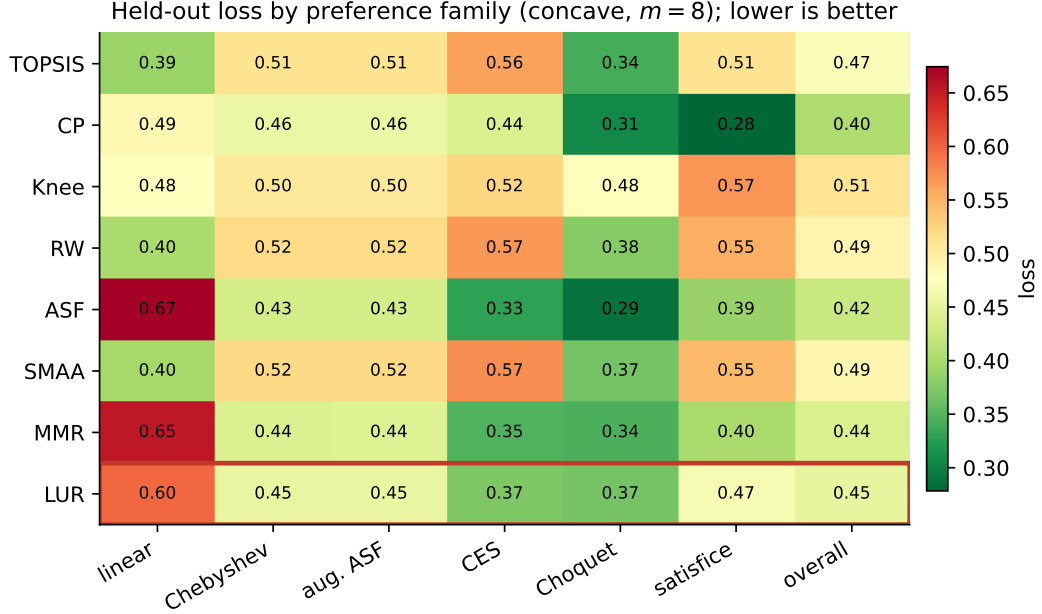


Figure 10: Held-out loss by preference family (concave, $m = 8$); greener is better. Averaging rules (TOPSIS, SMAA) are strong on linear utilities but weak on max-type families; ASF and MMR are the reverse. LUR (outlined) is the most balanced row across the additive *and* non-additive (Choquet, satisfice) families without being tuned to any.

Table 6: Redundant-objective benchmark (true preferences over the underlying criteria; 4 group structures $\times$ 3 geometries, 30 replications). Grouped loss measures de-redundification quality. Lower is better; best in **bold**.

Method	grouped loss	tail loss
TOPSIS	0.332	0.455
CP	0.317	0.423
Knee	0.322	0.489
RW	0.409	0.598
ASF	0.266	0.406
SMAA	0.413	0.603
MMR	0.295	0.387
LUR	0.268	0.391

to 0.420 but inflates certificate regret to 0.929, defeating the robustness aim; clustering is, by design, a no-op when criteria are independent (its value is the redundancy result); Figure 12 summarises the ablation, and Figure 13 the LUR–SMAA distinctness across geometries and m .

LUR is insensitive to the clustering threshold θ over $[0.3, 0.9]$ across all four geometries. Nadir perturbations reveal an important practical limitation. Although many changed winners remain close in held-out loss, the identity of the LUR recommendation can change frequently because the leximax certificate often contains near-ties. In the frozen audit protocol, the nadir-error stability gate was marked CHECK rather than PASS. Reliable normalisation bounds therefore remain a practical condition for using LUR, and users should report both winner-change rates and loss degradation under plausible bound perturbations. Table 7 and Figure 14 quantify these diagnostics.

7.8. Runtime scalability

LUR achieves its robustness without Monte-Carlo preference sampling. On our evaluation hardware, evaluating a candidate set instance ($N = 300$, $m = 8$) takes under 10 milliseconds for LUR, comparable to simple scalarizations (TOPSIS, CP) and roughly two orders of magnitude faster than evaluating robust Monte-Carlo baselines (SMAA, MMR) with 1,000 weight samples.

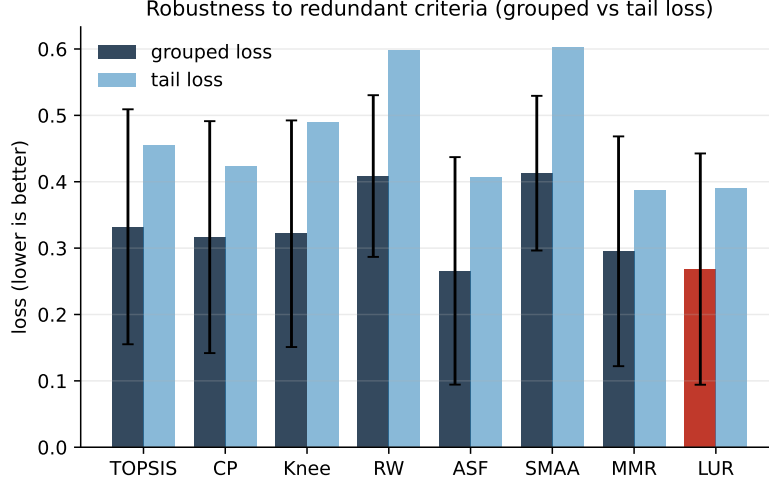


Figure 11: Robustness to redundant criteria. On grouped loss (true preferences over the underlying criteria) LUR ties ASF for best and clearly beats the averaging and Monte-Carlo rules, which over-weight correlated groups; the advantage is the reduced—not eliminated—redundancy sensitivity of the clustered probe family.

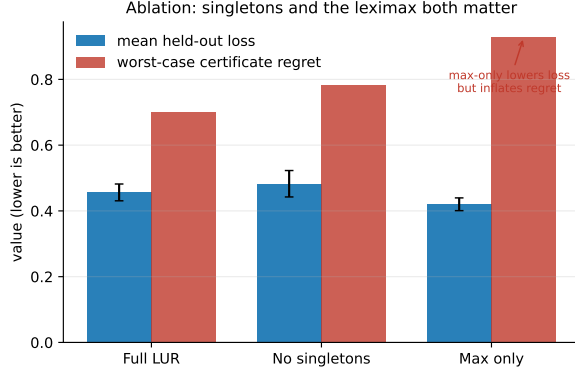


Figure 12: Ablation (concave, $m = 8$). Removing the singletons hurts both loss and certificate regret; a max-only family lowers mean loss but inflates worst-case certificate regret, defeating the robustness aim.

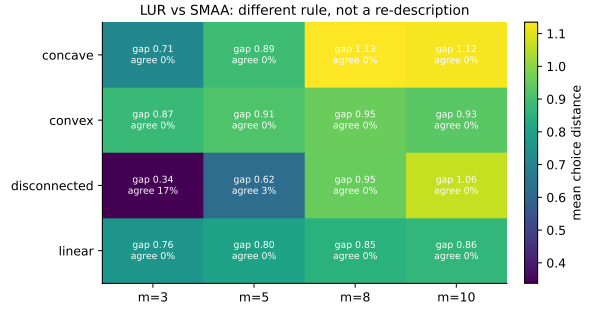


Figure 13: LUR versus the SMAA first-rank recommender: near-zero agreement and a large mean choice distance across geometries and m confirm LUR is a distinct rule, not a re-description of acceptability analysis.

7.9. Stochastic extension

Table 8 summarises the different stochastic metrics used to validate the extensions.

We validate the sample-size recovery (Proposition 2) directly. On noisy concave instances ($m = 6$) we draw n observations per criterion, form plug-in means, and apply LUR; Figure 15 reports how often the resulting choice matches the noise-free LUR winner. Recovery rises monotonically from 0.04 at $n = 1$ to 0.79 at $n = 100$, consistent with the consistency statement (it does not reach 1 because dense candidate sets contain near-ties, exactly the situation the tolerance rule is designed for).

7.10. Smart-grid dispatch case study

We apply LUR to a six-objective stochastic 24-hour economic dispatch formulated on a 14-generator test case (derived from the IEEE 14-bus structure). The objectives track total generation cost, CO₂ emissions, unreliability (loss of load probability), inter-hour ramping wear, non-renewable generation fraction, and the peak-to-average demand gap. The system is evaluated under Monte-Carlo renewable (wind/solar) generation scenarios. The 163 non-dominated dispatch policies were generated *a priori* using a standard NSGA-II evolutionary algorithm; this illustrates how LUR naturally interfaces with established multi-objective generation methods as a post-hoc robust selector without systematically

Table 7: Parameter diagnostics. Left: Held-out loss is highly stable to the clustering threshold θ . Right: Artificially corrupting the nadir estimate z^w by up to 30% flips the choice identity often due to near-ties, but worst-case quality (held-out loss) remains robust.

θ	linear	concave	convex	discon.	nadir err.	loss	flip rate
					0%	0.462	0%
0.3	0.179	0.459	0.036	0.351	5%	0.450	47%
0.5	0.179	0.452	0.036	0.355	10%	0.466	27%
0.8	0.176	0.452	0.037	0.360	20%	0.453	80%
					30%	0.446	93%

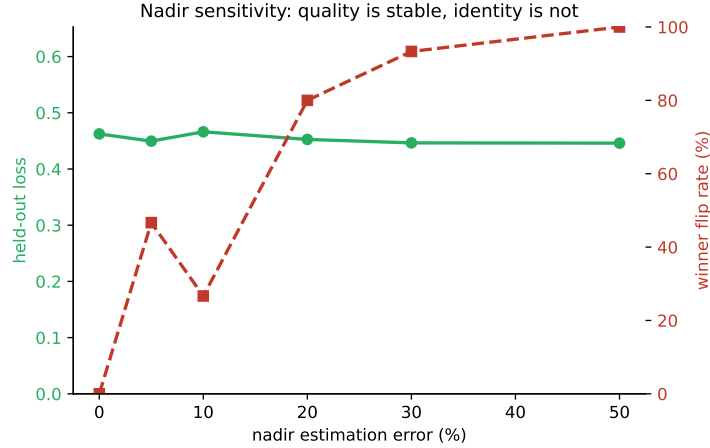


Figure 14: Nadir sensitivity. As the nadir estimate is corrupted, held-out *quality* stays essentially flat (green) while the *identity* of the winner flips frequently (crimson)—a direct consequence of the near-ties in the leximax certificate and of LUR’s set-dependence (§8). This motivates deploying LUR with conservative interval bounds.

biasing the candidate set. The full dispatch model (demand profile, generator limits, ramping, scenario generation) is formulated in Appendix A and fully specified in the released code. Relative to the distance/averaging methods, LUR accepts a negligible increase in *mean* held-out loss (0.103 vs 0.100) for a 23% reduction in *worst-case* loss (0.282 vs 0.366; knee 0.623). Crucially it *explains* the choice: the certificate (Figure 16) flags the peak-shaving gap (f_6 , regret 0.40) and the non-renewable fraction (f_5 , 0.40), then the ramping/peak coalition $\max(f_4, f_6)$ (0.36), telling the DM not only which dispatch to adopt but where it remains fragile.

8. Discussion and limitations

What LUR is, and is not.. LUR is a robust, certifiable decision rule. On worst-case held-out loss LUR is in the best-performing group—practically equivalent to achievement scalarisation and linear minimax regret (negligible effect size)—while on average-case loss the classical averaging methods do better. LUR’s distinctive value is the *combination* it offers that no single competitor matches: best-group worst-case robustness across additive *and* non-additive preferences, reduced sensitivity to redundant criteria, an interpretable certificate, and native stochastic and multi-stakeholder extensions. Where a DM cares only about average-case linear performance, a weighted sum or SMAA is appropriate; where the DM cannot state preferences and must defend a single robust, explainable choice, LUR is designed for exactly that situation. Table 9 provides a concise summary mapping the central claims of the paper to their theoretical or empirical evidence.

Relationship to existing rules.. LUR generalises rather than replaces. A single grand-maximum probe recovers the Chebyshev/ASF rule; a single grand-mean probe recovers the weighted sum; the singleton

Table 8: Summary of stochastic metrics reported across Sections 7.8 and 7.9.

Stochastic experiment	Metric	Compared methods	Result & Interpretation
Sample-size recovery	probability of matching noise-free LUR winner	stochastic LUR only	recovery rises with S; consistency diagnostic
Deterministic vs stochastic LUR	tail held-out loss	det. LUR vs stoch. LUR	0.383 vs 0.383; stochastic adjustment does not hurt tail loss
Out-of-sample risk	worst-criterion risk	stoch. LUR vs TOPSIS/SMAA	near 0 vs 0.28/0.52; confidence-aware certificate reduces risk

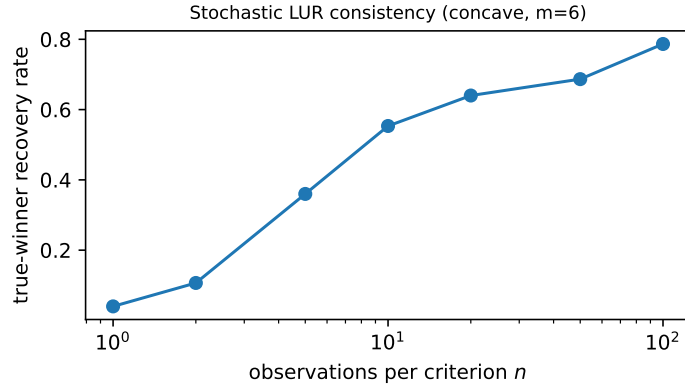


Figure 15: Stochastic-LUR consistency: recovery rate of the noise-free winner vs. the number of observations per criterion (concave, $m = 6$).

family recovers lexicographic maximin of normalised regret. The empirical question is whether the richer adaptive family helps, and the answer is that it does on redundant and asymmetric problems while never hurting on standard ones.

Independence of Irrelevant Alternatives (IIA) and Set-Dependence.. Because both the normalisation bounds (z^*, z^{nad}) and the probe anchors (q^*, q^w) are computed empirically over the active candidate set A , LUR does not satisfy Independence of Irrelevant Alternatives (IIA). Adding or removing a non-selected, non-dominated alternative that defines an extremum shifts the normalisation for *all* alternatives, which can change the sorted disappointment vectors and cause rank reversal. This is structurally the same vulnerability identified in TOPSIS (§1). This set-dependence explains the high winner-flip rates observed in our nadir-sensitivity experiments (Table 3). Consequently, while affine-rescaling invariance holds for a fixed set, LUR is highly sensitive to the composition of A . This underscores why deploying LUR with fixed, conservative *a priori* bounds (interval-robust LUR) is mandatory when stability against candidate-set perturbations is required.

The Certificate’s Decision Value.. While robust scalarisations like ASF provide similar worst-case guarantees, LUR’s distinct value lies in its auditable certificate. A standard sensitivity sweep over ASF reference points or a SMAA acceptability index reveals *how often* an alternative wins, but does not explain *why* it loses in specific contexts. In contrast, the LUR certificate explicitly isolates the binding constraints: it names the specific declared probe (e.g., “the average of cost and emissions”) that forces the maximum disappointment, giving the decision maker a direct semantic justification for the compromise that opaque distributional indices omit. Figure 17 shows the anatomy of such a certificate.

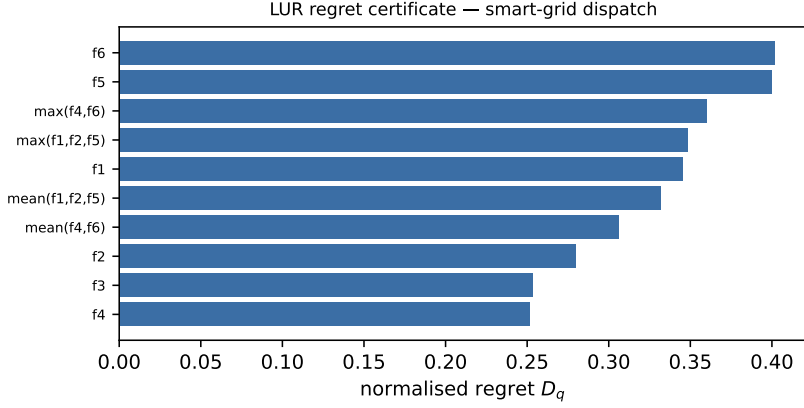


Figure 16: LUR regret certificate for the recommended smart-grid dispatch: the most disappointed probes (peak gap f_6 , non-renewable fraction f_5 , and the ramping/peak coalition) localise the fragility of the recommendation.

Table 9: Claim-evidence matrix mapping theoretical properties and empirical findings to their supporting evidence.

Claim	Evidence
Pareto compatibility	Theorem 2 (general) and Theorem 4 (non-negative probes)
Existence and completeness	Theorem 1
Tolerance-rule stability	Theorem 3 (margin condition)
Reduced redundancy bias	Table 6 (grouped loss)
Robustness on tail held-out loss	Figure 9 and Table 4
Stochastic consistency	Proposition 2 and Figure 15
Smart-grid applicability	Appendix A and Figure 16

Scope, Parameters, and Limitations.. LUR is a leximin-refined robust selector. It complements exploratory tools like SMAA by providing a single, auditable point recommendation when the DM must defend a robust choice. The framework relies on several explicit choices, summarised in Table 10. LUR’s contribution is that these choices are declared and separated from the final recommendation.

Table 10: Consolidated list of LUR parameters, their roles, and recommended defaults.

Param	Role	Default	Note
z^*, z^{nad}	Normalisation bounds	Empirical over A	Must use fixed <i>a priori</i> bounds if IIA/stability is required.
$\mathcal{Q}$	Probe family	Adaptive family	Defines the semantic interpretations of preference.
θ	Clustering threshold	0.6	Empirically insensitive over $[0.4, 0.8]$.
τ	Leximax tolerance	10^{-4}	Set to the scale of the disappointment estimation error.
ε	Strict domination margin	10^{-6}	Small fraction of the smallest non-degenerate criterion range.
α	Stochastic confidence	0.95	Used only in the stochastic extension (§6).

Threats to validity.. The held-out families, though disjoint from the probes, are themselves a modelling choice; we mitigate this by spanning linear, Chebyshev, ASF and CES forms and by reporting per-family results. Candidate sets are synthetic but cover eight audit geometries, with four canonical geometries used in the mechanistic study, and the dispatch illustration is also synthetic; extending to a wider library of real MCDA datasets is future work.

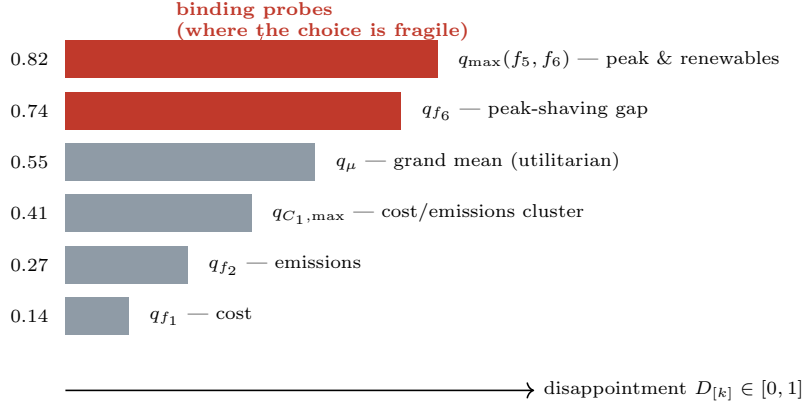


Figure 17: Anatomy of a regret certificate. The recommendation is reported with its probe disappointments sorted worst-first; the top, highlighted entries are the *binding probes* that drive the worst-case regret and localise the fragility of the choice (illustrative labels follow the smart-grid case of §7).

Interactive elicitation using certificates.. The certificate could support a future interactive workflow in which the DM reviews the binding probes and revises the declared family. For example, a DM might relax a cost-focused probe after seeing its effect on the recommendation, after which LUR can be recomputed. This possibility has not been evaluated here: removing a separating singleton can weaken Pareto-exclusion guarantees, so an interactive protocol would require explicit safeguards and user validation.

Future work.. Natural next steps include large-scale empirical validation of the continuous direct formulations (§4) on high-dimensional problems; and full development and evaluation of the exploratory Rawlsian multi-stakeholder variant. Finally, extending the family to include non-linear probes would significantly broaden LUR’s expressive power. For instance, capturing Cobb-Douglas structures (constant elasticity of substitution) could be achieved by evaluating linear probes in log-transformed criterion space, while modelling threshold-based satisficing preferences with interaction terms would require explicitly incorporating boolean conditionals into the anchor computations.

9. Conclusion

We have presented the Leximax Universal-Regret core, a Pareto-compatible decision rule that replaces the efficient set as the final object of multicriteria analysis with an auditable certificate. LUR operates as a leximin-refined preference-robust selection rule: it ranks alternatives by the lexicographically sorted vector of normalised achievement shortfalls over a declared family of monotone probes, and returns a robust stability class (often a singleton recommendation) together with an explanation of where it remains fragile. We proved existence and completeness, Pareto compatibility *under a separation condition*, dominated-solution exclusion, stability of the tolerance-based rule under bounded perturbations, and a disappointment-perturbation bound with explicit constants, and we gave a redundancy-aware, correlation-clustering probe construction that keeps the family linear in the number of criteria. A replicated held-out study over ten preference families and a stochastic smart-grid case showed that LUR attains best-group worst-case robustness—performing non-inferiorly to the strongest robust scalarisations—while reducing sensitivity to redundant criteria, certifying its choices, and extending naturally to noisy criteria.

The central message is that robust selection need not rely on exhaustive front generation or eliciting scalarisation weights. However, this robustness relies entirely on the reliability of the supplied normalisation bounds; if true nadirs are uncertain, the framework must be deployed conservatively using interval

estimates. By relocating preference choices into a declared class of questions, LUR turns a candidate set into a defensible, explainable recommendation—a low-disappointment compromise across the declared interpretations of the criteria, accompanied by a certificate that reports which questions remain most binding.

Reproducibility. All code, generated data, and scripts that regenerate every table and figure in this paper are released in an open-source repository (URL withheld to preserve anonymity during review; it will be provided in the camera-ready version), including the four-geometry benchmark, the redundancy benchmark, the statistical tests, the sensitivity analyses, and the smart-grid case study.

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Appendix A. Smart-grid economic dispatch formulation

The case study in §7 formulates a 24-hour stochastic economic dispatch with recourse on a 14-generator test case (derived from the IEEE 14-bus structure). The first-stage decision variables are the day-ahead unit commitment status u_{gt} . The second-stage (recourse) variables under scenario ω are the scheduled power $p_{gt}(\omega)$, renewable curtailment $c_t^{\text{ren}}(\omega)$, and load shedding $s_t^{\text{shed}}(\omega)$. The uncertain parameters are the demand $d_t(\omega)$, non-dispatchable renewable generation $w_t(\omega)$ (wind) and $s_t(\omega)$ (solar) under scenario ω , and line losses $\ell_t(\omega)$.

The power-balance constraint ensures that total generation meets demand and losses:

$$\sum_g p_{gt}(\omega) + w_t(\omega) + s_t(\omega) - c_t^{\text{ren}}(\omega) + s_t^{\text{shed}}(\omega) = d_t(\omega) + \ell_t(\omega), \quad \forall t, \omega. \quad (\text{A.1})$$

Generators are subject to minimum and maximum capacity bounds and ramping constraints R_g :

$$P_g^{\min} u_{gt} \leq p_{gt}(\omega) \leq P_g^{\max} u_{gt}, \quad |p_{g,t}(\omega) - p_{g,t-1}(\omega)| \leq R_g, \quad \forall \omega. \quad (\text{A.2})$$

The six objectives to be minimised are:

1. **Cost:** Expected total fuel cost.
2. **Emissions:** Expected total CO₂ emissions.
3. **Unreliability:** Loss of load probability $f_3(x) = \mathbb{P}(\sum_t s_t^{\text{shed}}(\omega) > 0)$.
4. **Ramping wear:** Expected absolute generation change $\mathbb{E}[\sum_{g,t} |p_{g,t}(\omega) - p_{g,t-1}(\omega)|]$.
5. **Non-renewable fraction:** Expected ratio of thermal to total generation.
6. **Peak stress:** Expected difference between the maximum and average dispatch-dependent net load on the thermal fleet.

The model is evaluated using a Monte-Carlo sample-average approximation over 100 scenarios. All parameter values (generator data, costs, emissions, ramp limits, and scenario bounds) are available in the open-source repository. This formulation serves as an illustrative application of the robust certificate framework.

Appendix B. Multi-stakeholder (Rawlsian) exploratory variant

Let $\mathcal{U} = \{1, \dots, U\}$ index stakeholders, each with importance weights $w_u \in \Delta^m$ inducing a stakeholder-specific disappointment $D_q^u(x)$. An exploratory *Rawlsian LUR* applies the leximax order to the worst-stakeholder disappointments, $\max_u D_q^u(x)$. It maintains natural fairness properties such as anonymity and worst-stakeholder protection. A full treatment of this extension is beyond the scope of this paper, but we include this preliminary formulation to demonstrate the flexibility of the regret-certificate framework.